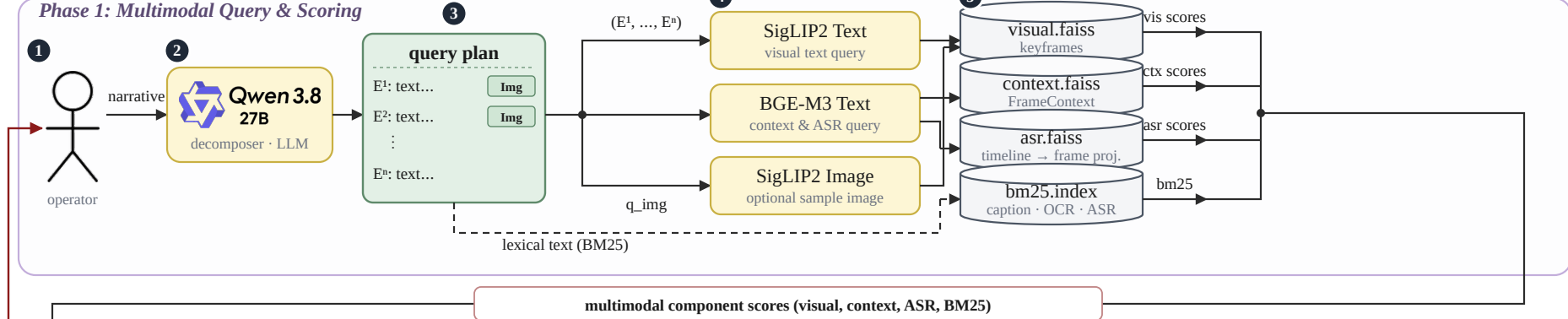
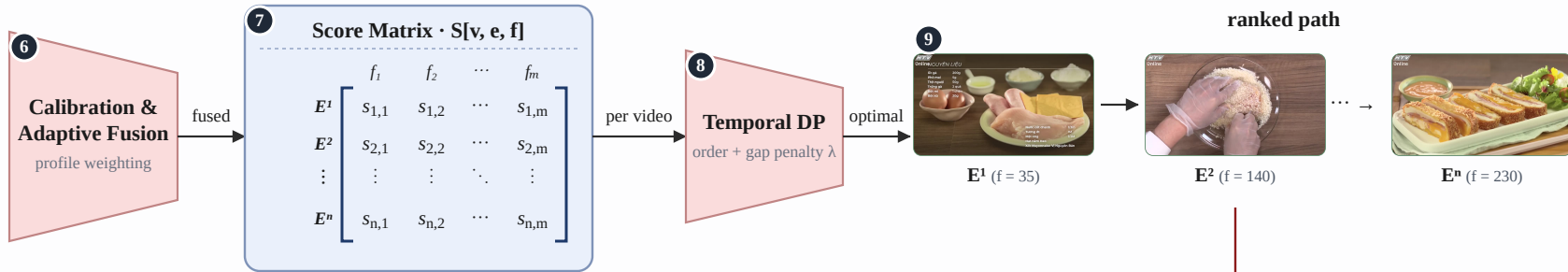


- ① Operator query → ② Qwen decomposes events → ③ Query plan → ④ Multimodal encoders (SigLIP2 & BGE-M3) → ⑤ Dense & BM25 retrieval → ⑥ Calibration & adaptive fusion → ⑦ Score matrix $S[v, e, f]$ → ⑧ Temporal DP aligns path → ⑨ Ranked paths & snapshot

Phase 1: Multimodal Query & Scoring



Phase 2: Calibration, Adaptive Fusion & Temporal Alignment



ranked candidate paths + evidence snapshot → operator browsing